

新增HsaeLocationProvider上报定位信息

一、framework方面修改

(1) 在framework的res.apk中新

增 config_enableHsaeLocationOverlay 和 config_hsaeLocationProviderPackageName 字段,

```
1 | <!-- frameworks/base/core/res/res/values/config.xml -->
2 | <bool name="config_enableHsaeLocationOverlay" translatable="false">true</bool>
3 | <string name="config_hsaeLocationProviderPackageName" translatable="false">com.example.lo
```

```
1 | <!-- frameworks/base/core/res/res/values/symbols.xml -->
2 | <java-symbol type="bool" name="config_enableHsaeLocationOverlay" />
3 | <java-symbol type="string" name="config_hsaeLocationProviderPackageName" />
```

(2) API新增 hsae_loc provider类型

frameworks/location

```
1 | public static final String HSAE_PROVIDER = "hsae_loc";
```

Tips: API新增后需要执行以下命令, 需要同步更新 framework/base/api/current.txt , 否则整编会抛出异常。

```
make update-api
```

(3) 在LocationMangerService启动时, 添加注册HsaeLocationService流程

```

1 // frameworks/base/services/core/java/com/android/server/location/LocationManagerService.
2 import static android.location.LocationManager.HSAE_PROVIDER;
3
4 public class LocationManagerService extends ILocationManager.Stub {
5
6     private static final String HSAE_LOCATION_SERVICE_ACTION =
7         "com.android.location.service.HsaeLocationProvider";
8
9     @GuardedBy("mLock")
10    private void initializeProvidersLocked() {
11        // bind to hsaeProvider
12        LocationProviderProxy hsaeProvider = LocationProviderProxy.createAndRegister(
13            mContext,
14            HSAE_LOCATION_SERVICE_ACTION,
15            com.android.internal.R.bool.config_enableHsaeLocationOverlay,
16            com.android.internal.R.string.config_hsaeLocationProviderPackageName);
17        if (hsaeProvider != null) {
18            LocationProviderManager hsaeManager = new LocationProviderManager(HSAE_PROVIDE
19            mProviderManagers.add(hsaeManager);
20            hsaeManager.setRealProvider(hsaeProvider);
21        } else {
22            Log.e(TAG, "no hsae location provider found");
23        }
24    }
25 }

```

二、创建HaseLocationProvider提供定位更新功能

(1) 创建HsaeLocationProvider

```
1 package com.example.locationprovider;
2
3 import android.content.Context;
4 import android.location.Criteria;
5 import android.location.Location;
6 import android.os.Handler;
7 import android.os.Message;
8 import android.os.SystemClock;
9 import android.os.WorkSource;
10
11 import com.android.location.provider.LocationProviderBase;
12 import com.android.location.provider.LocationRequestUnbundled;
13 import com.android.location.provider.ProviderPropertiesUnbundled;
14 import com.android.location.provider.ProviderRequestUnbundled;
15 import com.example.locationprovider.util.LogUtil;
16
17 public class HsaeLocationProvider extends LocationProviderBase {
18
19     private static final String TAG = "HsaeLocationProvider";
20
21     private float lat = 1.0f;
22     private float lon = 10.0f;
23
24     private static final ProviderPropertiesUnbundled PROPERTIES =
25         ProviderPropertiesUnbundled.create(
26             /* requiresNetwork = */ false,
27             /* requiresSatellite = */ false,
28             /* requiresCell = */ false,
29             /* hasMonetaryCost = */ false,
30             /* supportsAltitude = */ true,
31             /* supportsSpeed = */ true,
32             /* supportsBearing = */ true,
33             Criteria.POWER_LOW,
34             Criteria.ACCURACY_FINE
35         );
36
37     public HsaeLocationProvider(Context context) {
38         super(TAG, PROPERTIES);
39     }
40
41     private Handler mHandler = new Handler() {
42         @Override
43         public void handleMessage(Message msg) {
```

```
44         LogUtil.i(TAG, "handleMessage: reportLocation");
45         Location location = new Location("hsae_loc");
46         location.setLatitude(lat + 0.1);
47         location.setLongitude(lon + 0.2);
48         location.makeComplete();
49         reportLocation(location);
50         mHandler.sendMessageDelayed(1, 1000);
51     }
52 };
53
54 @Override
55 protected void onSetRequest(ProviderRequestUnbundled request, WorkSource source) {
56
57 }
58
59 public void startMock() {
60     LogUtil.i(TAG, "startMock");
61     mHandler.sendMessageDelayed(1, 1000);
62 }
63 }
64
```

```

package com.example.locationprovider;

import android.app.Service;
import android.content.Intent;
import android.os.IBinder;

public class HsaeLocationService extends Service {

    private HsaeLocationProvider mProvider;

    @Override
    public IBinder onBind(Intent intent) {
        if (mProvider == null) {
            mProvider = new HsaeLocationProvider(this);
            mProvider.startMock();
        }
        return mProvider.getBinder();
    }

    @Override
    public void onDestroy() {
        if (mProvider != null) {
            mProvider = null;
        }
    }
}

```

(2) 请求Location

```

package com.example.locationprovider;

import android.app.Activity;
import android.content.Context;
import android.location.Location;
import android.location.LocationListener;
import android.location.LocationManager;
import android.os.Bundle;

public class MainActivity extends Activity {
    private static final String TAG = "MainActivity";

    private LocationManager locationManager;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        initLocationManager();
    }

    private void initLocationManager() {
        Log.i(TAG, "initLocationManager");
        locationManager = (LocationManager) getSystemService(Context.LOCATION_SERVICE);
        locationManager.requestLocationUpdates("hsae_loc", 1000, 0, new LocationListener() {
            @Override
            public void onLocationChanged(Location location) {
                Log.i("onLocationChanged: ", " lat = " + location.getLatitude() + ", long
            }

            @Override
            public void onStatusChanged(String provider, int status, Bundle extras) {
                Log.i(TAG, "onProviderEnabled: onStatusChanged=" + provider);
            }

            @Override
            public void onProviderEnabled(String provider) {
                Log.i(TAG, "onProviderEnabled: provider=" + provider);
            }

            @Override
            public void onProviderDisabled(String provider) {
                Log.i(TAG, "onProviderDisabled: provider=" + provider);
            }
        });
    }
}

```

```

44         }
45     });
46 }
47
48 }

```

三、调试

上面的测试Demo中，HsaeLocationProvider 每隔1s就会上报一个位置信息，应用却无法收到 onLocationChanged 回调，查看日志发现被Blocking了：

```

04-25 15:35:42.239 800 2032 W LocationManagerService: blocking incomplete location from hsae_loc provider
04-25 15:35:43.241 26235 26235 I HsaeLocationProvider_Log: HsaeLocationProvider : handleMessage: reportLocation
04-25 15:35:43.241 800 2032 W LocationManagerService: blocking incomplete location from hsae_loc provider
04-25 15:35:44.243 26235 26235 I HsaeLocationProvider_Log: HsaeLocationProvider : handleMessage: reportLocation
04-25 15:35:44.245 800 2032 W LocationManagerService: blocking incomplete location from hsae_loc provider
04-25 15:35:45.246 26235 26235 I HsaeLocationProvider_Log: HsaeLocationProvider : handleMessage: reportLocation
04-25 15:35:45.246 800 2032 W LocationManagerService: blocking incomplete location from hsae_loc provider
04-25 15:35:46.248 26235 26235 I HsaeLocationProvider_Log: HsaeLocationProvider : handleMessage: reportLocation
04-25 15:35:46.249 800 2032 W LocationManagerService: blocking incomplete location from hsae_loc provider
04-25 15:35:47.250 26235 26235 I HsaeLocationProvider_Log: HsaeLocationProvider : handleMessage: reportLocation
04-25 15:35:47.251 800 2032 W LocationManagerService: blocking incomplete location from hsae_loc provider

```

分析源码可以看到

```

1  public void onReportLocation(Location location) {
2
3      // ...
4
5      // 在回调位置信息给客户端时，会有一个isComplete的判断，在这里卡住了
6      if (!location.isComplete()) {
7          Log.w(TAG, "blocking incomplete location from " + mName + " provider");
8          return;
9      }
10     // ...
11
12     handleLocationChangedLocked(this, location, mLocationFudger.createCoarse(loca
13 }

```

接着，查看一下Location中相关方法

```

1      @SystemApi
2      public boolean isComplete() {
3          if (mProvider == null) return false;
4          if (!hasAccuracy()) return false;
5          if (mTime == 0) return false;
6          if (mElapsedRealtimeNanos == 0) return false;
7          return true;
8      }
9
10     @TestApi
11     @SystemApi
12     public void makeComplete() {
13         if (mProvider == null) mProvider = "?";
14         if (!hasAccuracy()) {
15             mFieldsMask |= HAS_HORIZONTAL_ACCURACY_MASK;
16             mHorizontalAccuracyMeters = 100.0f;
17         }
18         if (mTime == 0) mTime = System.currentTimeMillis();
19         if (mElapsedRealtimeNanos == 0) mElapsedRealtimeNanos = SystemClock.elapsedRealti
20     }

```

调用makeComplete方法将location必要信息填满，再次运行测试Demo，HsaeLocationProvider上报的地址信息，可以正常回调给LocationListener了

```

04-25 15:43:41.300 28598 28598 I HsaeLocationProvider_Log: onLocationChanged: : lat = 1.1, long= 10.2
04-25 15:43:42.301 28598 28598 I HsaeLocationProvider_Log: HsaeLocationProvider : handleMessage: reportLocation
04-25 15:43:42.307 28598 28598 I HsaeLocationProvider_Log: onLocationChanged: : lat = 1.1, long= 10.2
04-25 15:43:43.308 28598 28598 I HsaeLocationProvider_Log: HsaeLocationProvider : handleMessage: reportLocation
04-25 15:43:43.313 28598 28598 I HsaeLocationProvider_Log: onLocationChanged: : lat = 1.1, long= 10.2
04-25 15:43:44.314 28598 28598 I HsaeLocationProvider_Log: HsaeLocationProvider : handleMessage: reportLocation
04-25 15:43:44.319 28598 28598 I HsaeLocationProvider_Log: onLocationChanged: : lat = 1.1, long= 10.2
04-25 15:43:45.320 28598 28598 I HsaeLocationProvider_Log: HsaeLocationProvider : handleMessage: reportLocation
04-25 15:43:45.322 28598 28598 I HsaeLocationProvider_Log: onLocationChanged: : lat = 1.1, long= 10.2
04-25 15:43:46.324 28598 28598 I HsaeLocationProvider_Log: HsaeLocationProvider : handleMessage: reportLocation
04-25 15:43:46.329 28598 28598 I HsaeLocationProvider_Log: onLocationChanged: : lat = 1.1, long= 10.2
04-25 15:43:47.330 28598 28598 I HsaeLocationProvider_Log: HsaeLocationProvider : handleMessage: reportLocation
04-25 15:43:47.334 28598 28598 I HsaeLocationProvider_Log: onLocationChanged: : lat = 1.1, long= 10.2

```